Body mass, growth and social-communication phenotypes in carriers of 16p11.2 and 22q11.2 CNVs: manuscript sections

Draft sections for the Nature Genetics manuscript. Part 1 (Results section on body mass, with the Figure 6 legend) and the Figure 6 Methods in Part 2 were re-synced on 2026-10-27 from the master Google Doc, after the Fig. 6 prime panels became age-band bars; the Supplementary Information in Part 3 is the 2026-10-26 draft and its numbering may lag the Google Doc, which is the master copy.

Part 1. Main text

BMI is a modifier of growth and autistic traits with effects that vary between CNVs

Childhood obesity has an influence on height trajectories in the population (Chung 2017) and in rare genetic disorders (Martinelli et al. 2011; Zufferey et al. 2012); however, gene x BMI interactions on growth have not been studied systematically. Several CNVs in this study have effects on both height and BMI (15q11.2 BP1-BP2, 16p11.2 BP2-BP3, 16p11.2 BP4-BP5, 17q12). Effects of CNVs on growth could therefore be intertwined with the modifying effects of metabolism. However

We analysed carriers of 16p11.2 (BP4-BP5) and 22q11.2 (A-D) deletions and duplications from the G2MH, Simons Searchlight, Simons VIP and Cardiff cohorts together with adult carriers from the UK Biobank (height: 507, 372, 542 and 406 carriers of 16p11.2 deletion, 16p11.2 duplication, 22q11.2 deletion and 22q11.2 duplication, of whom 117, 149, 5 and 270 were UK Biobank adults; head circumference, clinical cohorts only: 230, 142, 339 and 106). Non-carrier controls were unaffected relatives from the Simons Simplex Collection (SSC) [PMID:20955926, to add] and Simons VIP48 (n = 6,293) and, for height, 50,000 randomly drawn UK Biobank non-carriers. Children were scored against external growth references and adults relative to the non-carrier adults of their own cohort (Methods). We modelled each outcome with generalized additive models in which the BMI effect is the age-adjusted slope of the outcome on BMI z (s.d. per s.d.), and compared carriers with controls in the difference of that slope (CNV x BMI; **Fig. 6** and **Supplementary Table 22**). Mean height in carriers differed from controls in ways that changed over development: 16p11.2 deletion carriers were 0.32 s.d. shorter than controls in childhood and 1.06 s.d. shorter in adulthood, 16p11.2 duplication carriers were 0.63 s.d. shorter in childhood and 0.39 s.d. taller as adults, 22q11.2 deletion carriers were short at all ages (-0.99 and -0.53 s.d.), and 22q11.2 duplication carriers were slightly tall as children and slightly short as adults (+0.18 and -0.28 s.d.; **Fig. 6A-E**, **Supplementary Table 22**).

The effect of body mass on growth differed by genotype. In a pooled model of controls and the four CNV groups, the BMI slope differed among the five groups for both height (χ^2 = 14.4, 4 d.f., P = 0.006) and head circumference (χ^2 = 21.1, 4 d.f., P = 3.0 x 10^{-4}; **Supplementary Table 22**). In controls, BMI was positively associated with height in childhood and weakly negatively in adulthood (**Fig. 6A'**; overall -0.07 s.d. per s.d.). For height, the difference came from two CNVs in which the BMI slope was more negative than in controls: 16p11.2 duplication (CNV x BMI -0.13, P = 0.003) and 22q11.2 deletion (-0.09, P = 0.016). In 16p11.2 deletion and 22q11.2 duplication carriers, the slope did not differ from controls (-0.01, P = 0.80, and +0.05, P = 0.28). The difference was confined to development (five-group test at 2-9 years P = 8.3 x 10^{-5} and at 10-18 years P

= 0.029; in adults *P* >= 0.21): in 16p11.2 duplication carriers aged 2-9 years, higher BMI was associated with shorter stature (**Fig. 6C'**; -0.39 versus +0.12 s.d. per s.d. in controls, $P = 2.5 \times 10^{-6}$), and 22q11.2 deletion carriers lacked the positive association seen in adolescent controls (**Fig. 6D'**; -0.04 versus +0.14 at 10-18 years, *P* = 0.003; **Supplementary Table 21**). In 22q11.2 duplication carriers, BMI was positively associated with height in young adults, unlike in controls (+0.16 versus -0.08 s.d. per s.d. at 19-39 years, *P* = 0.027; **Fig. 6E'**).

For the 16p11.2 deletion, body mass in childhood rather than in adulthood was related to adult stature. In UK Biobank, adult carriers who recalled being plumper than average at age 10 were shorter (-0.28 s.d. per category, *P* = 0.036; non-carriers -0.07 s.d.; difference *P* = 0.10), whereas adult BMI was unrelated to their height (-0.03, *P* = 0.85; **Supplementary Fig. 5**, **Supplementary Table 24**). In longitudinal series from children, higher BMI at a visit preceded faster height growth to the next visit before 8 years of age (**Supplementary Fig. 1**, **Supplementary Table 13**). Together these observations are consistent with early-life obesity accelerating growth in childhood at the cost of adult height in 16p11.2 deletion carriers. In 16p11.2 duplication carriers the pattern was reversed: adult BMI, not childhood body size, was associated with shorter adult stature (-0.35 s.d. per category, *P* = 0.005; difference from non-carriers *P* = 0.043).

Head circumference rose with BMI in controls (+0.30 s.d. per s.d.), and the five-group difference came from 16p11.2 deletion carriers, who showed a stronger coupling of head size to body mass than controls (CNV x BMI +0.19 s.d. per s.d., $P = 6.6 \times 10^{-5}$; **Fig. 6G**), concentrated in early childhood (**Fig. 6G'**; +0.49 versus +0.25 s.d. per s.d. at 2-9 years, *P* = 0.001; **Supplementary Table 21**), whereas no other CNV differed from controls (*P* = 0.23-0.33). In children with repeated measurements, higher BMI preceded faster head growth (pooled carriers +0.10 s.d. per s.d., $P = 6.7 \times 10^{-4}$; **Supplementary Fig. 1**). Among carriers of the overgrowth CNVs (16p11.2 deletion and 22q11.2 duplication), those with autism had larger heads than those without (+0.45 s.d., *P* = 0.022), but BMI did not differ by autism status in any CNV group (**Supplementary Fig. 4**, **Supplementary Tables 17** and **18**).

Social-communication impairment also tracked body mass in carriers. Social Responsiveness Scale (SRS) [PMID:12959421, to add] and Social Communication Questionnaire (SCQ) [PMID:10789276, to add] scores rose with BMI in carriers, with slopes exceeding those of unaffected relatives for SCQ in all four CNVs (CNV x BMI +0.26 to +0.48 s.d. per s.d., *P* = 0.0010-0.034) and for SRS in three (+0.17 to +0.51, $P = 4.7 \times 10^{-6}$ to 0.047; 22q11.2 duplication -0.06, *P* = 0.58; **Fig. 6K-T**). Pooled across CNVs the excess was +0.22 s.d. per s.d. for SRS ($P = 1.2 \times 10^{-6}$) and +0.34 for SCQ ($P = 5.0 \times 10^{-5}$), with no heterogeneity between CNVs (*P* = 0.61 and 0.96). Neither unaffected relatives (+0.01) nor autistic SSC probands without a CNV (+0.03; *P* versus relatives 0.42 and 0.79) showed a BMI slope, so the association is not a general feature of autism; within carriers it was weaker in those without an autism diagnosis (pooled SRS +0.11 s.d. per s.d.; **Supplementary Table 20**). Cognitive performance (Penn CNB) was unrelated to BMI (**Supplementary Figs. 2** and **3**).

In the clinical cohorts, these associations were not explained by sex, polygenic background or medication. Adjusting the CNV x BMI contrasts for sex and for polygenic scores for childhood obesity, adult BMI, height and autism, singly or together, left the estimates unchanged, and the effects were present within each sex and within both halves of every polygenic-score distribution (**Supplementary Tables 14** and **15**). Psychotropic medication (29-33% of carriers) was associated with the traits but not with BMI, and the slopes were unchanged after adjustment for medication class or

restriction to unmedicated carriers (**Supplementary Table 19**). Together, these results show that body mass modifies the developmental effects of these CNVs in a CNV-specific way: it is coupled to early head growth in 16p11.2 deletion carriers, it is associated with growth restriction in young 16p11.2 duplication carriers and loses its adolescent growth association in 22q11.2 deletion carriers, and it tracks the severity of social-communication impairment across CNVs, in ways not seen in the general population or in idiopathic autism.

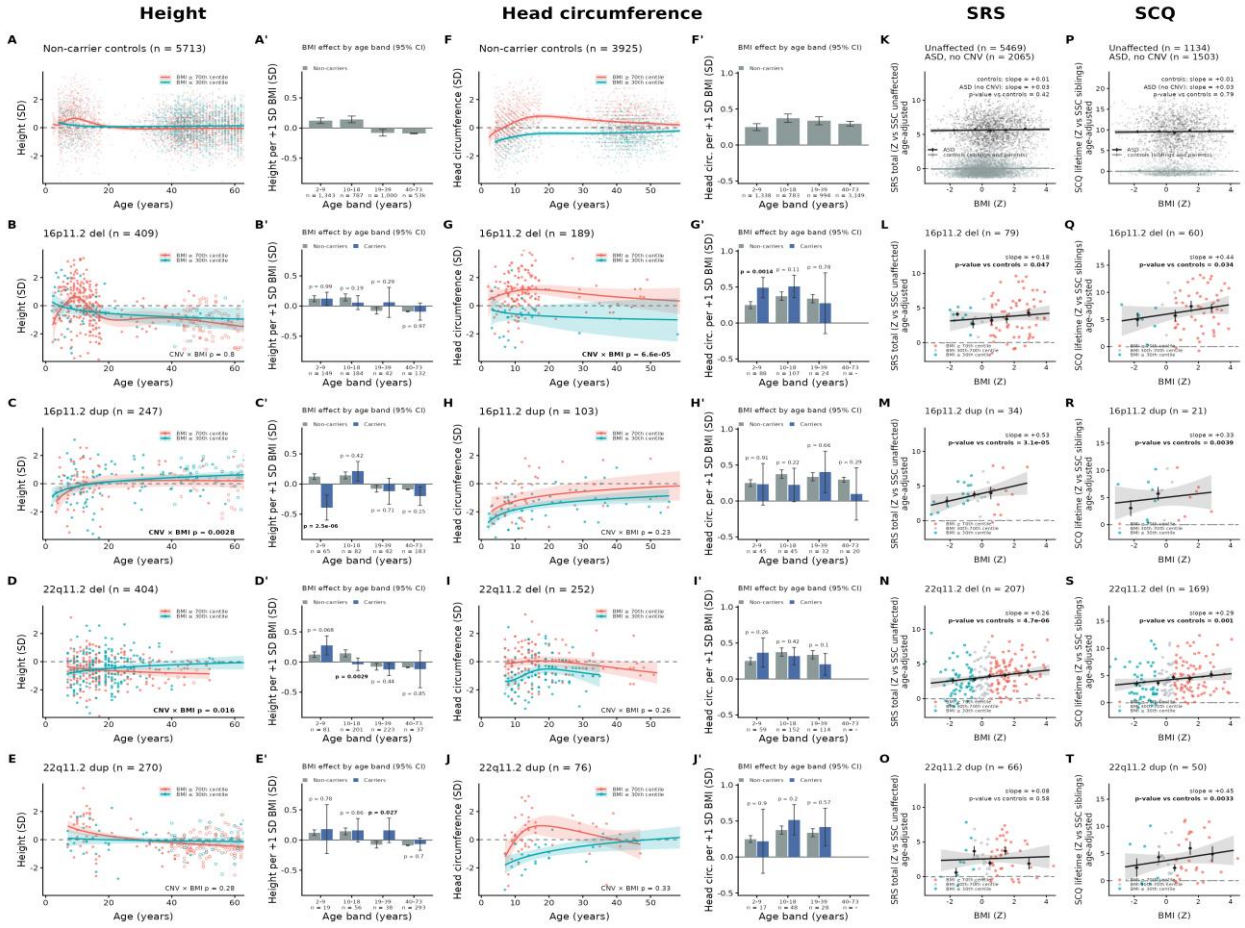


Figure 6. Body mass modifies the effects of 16p11.2 and 22q11.2 CNVs on growth and autistic traits.

A–E, Height *z* score against age in non-carrier controls (**A**) and in carriers of 16p11.2 deletion (**B**), 16p11.2 duplication (**C**), 22q11.2 deletion (**D**) and 22q11.2 duplication (**E**), for individuals with BMI at or above the 70th centile (red) or at or below the 30th centile (teal); clinical cohorts (filled circles) and UK Biobank adults aged 40–73 years (open circles); lines are generalized additive model (GAM) smooths with 95% confidence bands. The control panel shows the clinical non-carriers (SSC designated siblings and parents, Simons VIP non-carrier family members) and a random 3,000 UK Biobank non-carriers. **F–J**, Head-circumference *z* score in the same groups, clinical cohorts only. Printed values are the two-sided *P* value of the age-adjusted difference between carriers and controls in the BMI slope (CNV × BMI; $y \sim \text{CNV} + s(\text{age, by CNV}) + s(\text{age, by BMI}) + \text{CNV} \times \text{BMI}$, continuous BMI, fitted to all individuals, not only the plotted tails). **A′–J′**, The effect of BMI on the outcome within age bands (2–9, 10–18, 19–39 and 40–73 years; s.d. of outcome per s.d. of BMI, 95% confidence interval) in controls (grey) and carriers (blue), from one least-squares model per CNV and band, $y \sim \text{CNV} \times \text{BMI} + \text{sex} + \text{age}$; printed values are the two-sided P value of the carrier–control difference

(CNV × BMI), and the number under each band is the number of carriers (controls in A′ and F′). Statistics in Supplementary Table 21. **K–T**, SRS total (*z* relative to SSC unaffected relatives; **K–O**) and SCQ lifetime total (*z* relative to SSC siblings; **P–T**), adjusted for age within group, against BMI *z*; points coloured by BMI tail (grey, 30th–70th centile); black line and band, linear fit with 95% CI; diamonds, mean ± s.e.m. within BMI bins; dashed grey line, the fit in unaffected relatives; **K** and **P** also show SSC probands with autism and no CNV (black). Printed values are the within-group BMI slope and the two-sided *P* value of its difference from unaffected relatives. Children were scored against the WHO (height, BMI) and UK 1990 (head circumference) references; adults against the non-carrier adults of their own cohort (Box-Cox transformation with the UK Biobank exponent, adjusted for sex, age and age squared). The BMI slope in controls, each CNV's difference from it, and the test of whether the BMI effect differs among the five genotype groups are given in Supplementary Table 22. Sample sizes for the tests: height, 507, 372, 542 and 406 carriers (16p11.2 deletion, 16p11.2 duplication, 22q11.2 deletion, 22q11.2 duplication; 117, 149, 5 and 270 from UK Biobank) and 6,293 clinical plus 50,000 UK Biobank controls; head circumference, 230, 142, 339 and 106 carriers and 6,264 controls; SRS, 79, 34, 207 and 66 carriers, 5,469 relatives and 2,065 probands; SCQ, 60, 21, 169 and 50 carriers, 1,134 siblings and 1,503 probands. Bold, nominal $P < 0.05$; no adjustment for multiple comparisons.

Part 2. Methods (main figure)

Effects of BMI on growth and autistic traits

Participants. Carriers of 16p11.2 proximal (BP4-BP5) and 22q11.2 (LCR A-D) deletions and duplications were drawn from the Genes to Mental Health (G2MH) network, Simons VIP48, Simons Searchlight and the Cardiff University cohort [to confirm: inclusion criteria and consent statement per site], and from UK Biobank (participants of European genetic ancestry who passed the genotype quality control of the main analysis, with CNVs called as described above; one record per person, ages 40-73 years). Clinical controls were non-carrier designated siblings and parents from the Simons Simplex Collection (SSC) [PMID:20955926, to add] and non-carrier family members without autism from Simons VIP; for height, a random sample of 50,000 UK Biobank non-carriers was added. SSC probands with autism and no CNV formed a separate autism comparison group. Clinical participants were aged 2-73 years; for each individual we used the most recent visit at which the outcome and BMI were both available. Autism diagnosis in carriers was taken from the G2MH subject table and the Simons VIP diagnostic summary. Social-communication measures were the SRS-2 total raw score [PMID:12959421, to add] (parent report for children, adult self or informant report) and the SCQ lifetime total [PMID:10789276, to add].

Anthropometric measures. Height, weight and head circumference were taken from the records of each cohort and screened for implausible values (Supplementary Methods). Children (height and BMI below 19 years, head circumference below 18 years) were scored against the WHO child growth standards and the WHO 2007 reference [PMID:16817681, to add] [PMID:18026621, to add] and, for head circumference, the UK 1990 reference [PMID:9496720, to add]. Adults were scored relative to the non-carrier adults of their own cohort, as for the biobank analyses: within sex, the outcome was transformed with a Box-Cox exponent estimated once per trait and sex in the UK Biobank non-carriers (height 0.65 in men and 0.60 in women; BMI -0.75

and -1.05) and applied in every cohort, regressed on age and age squared in that cohort's non-carriers, and expressed as the residual divided by the non-carrier residual s.d. The clinical cohorts were treated as one cohort with the SSC parents and Simons VIP non-carrier relatives as reference; adult head circumference, which UK Biobank lacks, was not transformed. Non-carrier adults therefore have mean 0 in every cohort, and adult carrier values are read relative to non-carriers of the same cohort. SRS and SCQ totals were expressed as *z* scores relative to SSC unaffected relatives and SSC siblings, respectively.

Statistical analysis. Models were generalized additive models (GAMs) fitted by restricted maximum likelihood in mgcv [Wood 2011, J. R. Stat. Soc. B 73, 3-36; no PMID, to add] (R 4.6.1); BMI entered every model as a continuous *z* score, and the 70th and 30th centile split in Fig. 6 is for display only. Carriers were compared with controls in $y \sim$ CNV + s(age, by = CNV) + s(age, by = BMI) + CNV:BMI, in which the BMI effect may vary with age in a way shared by both groups and CNV:BMI (CNV x BMI) is the age-adjusted difference between carriers and controls in the BMI slope. To test whether the BMI effect differs by genotype, the five groups (controls and the four CNVs) were fitted together, $y \sim$ group + s(age, by = group) + s(age, by = BMI) + sum over the four CNVs of CNV:BMI, and the four CNV:BMI terms were tested jointly (4-df Wald test); heterogeneity among the four CNVs alone was tested within carriers in $y \sim$ s(age, by = locus) + locus + BMI + locus:BMI. Carrier mean shifts were estimated separately in children and adults from $y \sim$ CNV + s(age). The BMI effect within age bands (Fig. 6A'-J'; 2-9, 10-18, 19-39 and 40-73 years) was estimated by least squares in carriers and controls together, y ~ CNV x BMI + sex + age, separately for each CNV and band: the control slope is the BMI term, the carrier slope the sum of the BMI and CNV x BMI terms, and the printed P value tests the CNV x BMI term. Whether the BMI effect differed among the five groups within a band was tested by an F test of the four CNV x BMI terms in the same model fitted to all groups (Supplementary Tables 21 and 22). Smooth age-varying curves were not used because they proved sensitive to the choice of model and smoothing. For SRS and SCQ, whose BMI effects did not depend on age, the BMI slope was estimated in $y \sim$ BMI + s(age) within each group and compared with unaffected relatives in the model above. All tests were two-sided and *P* values are reported without adjustment for multiple comparisons. No statistical method was used to predetermine sample size.

Sensitivity and supporting analyses. Robustness of the carrier-control contrasts to sex, polygenic background and psychotropic medication, the longitudinal (prospective) analyses, the comparison of carriers with and without autism, the cognitive analyses and the UK Biobank analysis of recalled body size at age 10 are described in the Supplementary Methods.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article [to be completed].

Data availability

SSC and Simons VIP/Searchlight phenotype data are available to approved researchers through SFARI Base (https://base.sfari.org). G2MH phenotype data are available through the G2MH network [to confirm: NDA collection or data-access route]. Participant-level growth, phenotype and polygenic-score tables used here contain identifiers and are held in the controlled analysis data folder; summary statistics for every test are provided in Supplementary Tables 1-10.

Code availability

All analysis code, figure scripts, summary tables and the analysis report are available at https://github.com/jsebat/cnv-growth-bmi (scripts 00-41; the main figure, Fig. 6, is produced on Expanse by 35_main_figure_v5_ukbb.R, which sources 17_main_with_ukbb.R).

Part 3. Supplementary Information

Supplementary Results

Prospective effects of body mass on growth

In carriers under 18 years with repeated measures, higher BMI at a visit preceded faster height growth to the next visit only in early childhood: in 16p11.2 deletion carriers the age-varying effect differed from zero ($P = 0.006$), the all-ages coefficient was +0.09 s.d. of height *z* per s.d. of BMI *z* (s.e. 0.04, $t = 2.1$, $P = 0.036$) and the effect was concentrated below 8 years (pooled carriers, < 8 y window +0.13, $t = 2.9$, $P = 0.005$; 8-13 y +0.01; 13-18 y -0.07). For head circumference the prospective effect was positive at all ages (16p11.2 deletion +0.08, $t = 2.1$, $P = 0.038$; pooled carriers +0.10, $t = 3.4$, $P = 6.7 \times 10^{-4}$; age-varying $P = 7.5 \times 10^{-4}$) (Supplementary Fig. 1, Supplementary Table 3).

Autism diagnosis within CNV group

Head circumference was larger in carriers with than without an autism diagnosis in the overgrowth CNVs (pooled 16p11.2 deletion and 22q11.2 duplication: +0.45 s.d., $P = 0.022$, $n = 188$ without and 96 with autism; 16p11.2 deletion +0.41, $P = 0.074$; 22q11.2 duplication +0.59, $P = 0.16$) but not in the undergrowth CNVs (-0.05, $P = 0.8$); SSC probands also had slightly larger heads than their siblings (+0.14, $P = 0.0014$). Tested as an interaction, the autism effect in the overgrowth CNVs exceeded the proband-sibling difference (+0.35 s.d., s.e. 0.17, $P = 0.046$). Height did not differ by autism status in any group (all $P > 0.07$), and BMI did not differ by autism status within any CNV group (per locus +0.10 to +0.16 s.d., $P = 0.43$-0.78; interaction with carrier status $P > 0.6$ throughout), whereas SSC probands were marginally heavier than their siblings (+0.11, $P = 0.032$). The BMI effects in Fig. 1 are therefore not a by-product of autistic carriers being heavier. Adding polygenic scores for childhood obesity, adult BMI, height and autism, singly or together, to the age- and sex-adjusted models left all of these contrasts unchanged (Supplementary Fig. 4, Supplementary Tables 7 and 8).

Cognitive performance

Penn Computerized Neurocognitive Battery[13,14] accuracy and speed, available for G2MH carriers only, showed no reproducible association with BMI for any subtest or total (accuracy total: 16p11.2 deletion +0.02, 16p11.2 duplication -0.14 ($P = 0.074$), 22q11.2 deletion -0.02, 22q11.2 duplication +0.06 s.d. per s.d.; heterogeneity across CNVs $P = 0.11$). (Supplementary Figs. 2 and 3, Supplementary Table 2).

Sex and polygenic background

In the subset with polygenic scores (G2MH carriers and SSC controls), the CNV x BMI contrasts were essentially unchanged by adjustment for sex, for each polygenic score with its BMI interaction, for all four scores, or for all four plus sex, on the same individuals: SRS, all CNVs, +0.192 (s.e. 0.048, $P = 5.8 \times 10^{-5}$) unadjusted and +0.198 ($P = 3.6 \times 10^{-5}$) fully adjusted; SCQ, all CNVs, +0.318 ($P = 5.0 \times 10^{-4}$) and +0.312 ($P = 7.1 \times 10^{-4}$); head circumference, 16p11.2 deletion, +0.130 ($P = 0.086$; weaker than in the full sample because Simons VIP, Searchlight and Cardiff carriers are unscored) and +0.151 ($P = 0.041$) (Supplementary Table 5). The carrier-versus-control slopes were present in both sexes and in both halves of every polygenic-score split (Supplementary Table 4). Of

206 modifier x BMI interaction tests, 47 had at least 80% power to detect an interaction as large as the carriers' own BMI effect; eleven were nominally significant (about two expected by chance, although the tests are correlated) and none survived correction for 47 tests (smallest $P = 0.017$). The only directionally consistent signal was a stronger BMI-growth coupling with higher adult-BMI polygenic score (height, all CNVs, +0.088 per s.d. of score, $P = 0.017$; head circumference +0.089, $P = 0.031$) (Supplementary Table 6).

Psychotropic medication

Current psychotropic medication was recorded for 85-92% of G2MH carriers and used by 29-33% (antipsychotics 5-7%). Medication classes were strongly associated with the traits (antipsychotic users: SRS +2.3 s.d., SCQ +3.5 s.d., CNB accuracy -0.6 s.d.) but not with BMI (antipsychotics -0.2 s.d., $P = 0.5$; any psychotropic +0.3, $P = 0.09$), and the within-carrier BMI slopes were unchanged by adjustment for medication classes or by restriction to unmedicated carriers: SRS, all CNVs, +0.22, +0.19 and +0.23; SCQ +0.35, +0.37 and +0.38; head circumference, 16p11.2 deletion, +0.47, +0.48 and +0.50; height, 16p11.2 deletion, +0.31, +0.32 and +0.27 (Supplementary Table 9).

Supplementary Methods

Ages of SSC relatives

The age fields of the SSC version 15 phenotype tables give age at a single snapshot date (September 2012), about three years after the typical study visit. Visit age for siblings and parents was reconstructed as the proband's age at evaluation plus the difference in birth month between the relative and the proband; this reproduces the SSC's own tabulated sibling height z scores ($r = 0.96$).

Plausibility screen

Heights recorded in metres were multiplied by 100. Heights outside 60-230 cm (together with the derived BMI), BMI outside 10-70 kg m^{-2}, |height z| > 5, child |BMI z| > 5 and head circumference outside 30-70 cm or $|z| > 5$ were set to missing before any model was fitted. The screen was applied to the cross-sectional genomatrix and to the longitudinal Simons VIP series. Flagged records were listed for manual review and a correction sheet is applied automatically when filled in.

Longitudinal models

In carriers under 18 years, the change in height z or head-circumference z from one visit to the next was regressed on BMI z at the first visit, adjusting for z at that visit, interval length, sex and, in pooled models, locus and cohort, with a child random intercept. A mixed GAM with s(age, by = BMI) gave the age-varying effect and its test against zero at any age; the same model with a constant BMI term gave the all-ages coefficient; linear mixed models with BMI x age-window terms gave window-specific coefficients (height: < 8, 8-13, 13-18 years; head circumference: 2-5, 5-18 years). Height used longitudinal visits of Simons VIP, Searchlight and Cardiff carriers (intervals >= 0.5 years); head circumference used the full Simons VIP measurement series plus Cardiff, G2MH and SSC carrier visits. 22q11.2 duplication children were too few ($n <= 12$) for separate estimates.

Autism diagnosis within CNV group

Within each CNV group, the pooled overgrowth and undergrowth groups and the SSC (probands versus designated siblings), height, head-circumference and BMI z were compared between individuals with and without an autism diagnosis in GAMs $y \sim$ autism + s(age) + sex, plus cohort and, in pooled groups, locus. The difference between the autism effect in a carrier group and that in SSC non-carriers was tested by pooling

the carrier group with the non-carriers and fitting $y \sim$ autism + locus + autism x carrier + s(age, by = carrier) + sex + cohort. Sensitivity models added each polygenic score, and all four, in the subset with scores.

Polygenic scores

Weights were SBayesRC posterior effects[15] (UK Biobank European HapMap3 LD reference, baseline-LD 2.2 annotations) derived from published summary statistics for adult BMI[16], childhood obesity[17] and height[18], and from the April 2026 autism GWAS meta-analysis (iPSYCH, PGC, SPARK, SSC and MoBa; unpublished; the 2019 release is ref. 19). Scores were computed with PLINK 2[20] on whole-genome sequence genotypes of G2MH carriers and on the SSC genotype set; for autism, SSC was scored with weights derived leaving the SSC out of the discovery GWAS, because the SSC contributes to that GWAS. Scores were residualized on ten ancestry principal components within cohort and standardized within cohort. Simons VIP, Searchlight and Cardiff carriers were not scored.

Sex and polygenic-score sensitivity analyses

Confounding (Supplementary Table 5): for each trait and group, the carrier BMI slope and the CNV x BMI contrast versus unaffected controls were refitted on the individuals with all four scores, unadjusted; adding sex and sex x BMI; adding each score with its BMI interaction; adding all four; and all four plus sex. Stratification (Supplementary Table 4): the main-figure tests were repeated within sex and within below- and above-median halves of each score, carriers and controls stratified alike. Interactions (Supplementary Table 6): for each modifier (sex; each score as a continuous z), BMI x modifier was fitted within carriers ($y \sim$ s(age) + BMI x modifier [+ locus]); power to detect an interaction as large as the carriers' own BMI effect was computed from the term's standard error (normal approximation) and interactions are reported only where that power was at least 80%.

Medication

Current medication from the G2MH medications release was classified by drug name into antipsychotics, stimulant and other ADHD medication, ADHD alpha-agonists, antidepressants, antiepileptics and mood stabilizers, and anxiolytics; weight-gain-associated (antipsychotics, valproate, lithium, mirtazapine, gabapentinoids) and weight-loss-associated (stimulants, atomoxetine, topiramate, metformin, GLP-1 agonists) groupings were also formed. Within-carrier BMI slopes for height, head circumference, SRS, SCQ and CNB were refitted unadjusted, with class indicators, in carriers on no psychotropic, and with classes plus autism diagnosis. Associations of each class with BMI and with the trait were tested in age- and locus-adjusted GAMs. Controls have no medication data, so the carrier-control contrasts cannot be medication-adjusted.

Cognitive measures

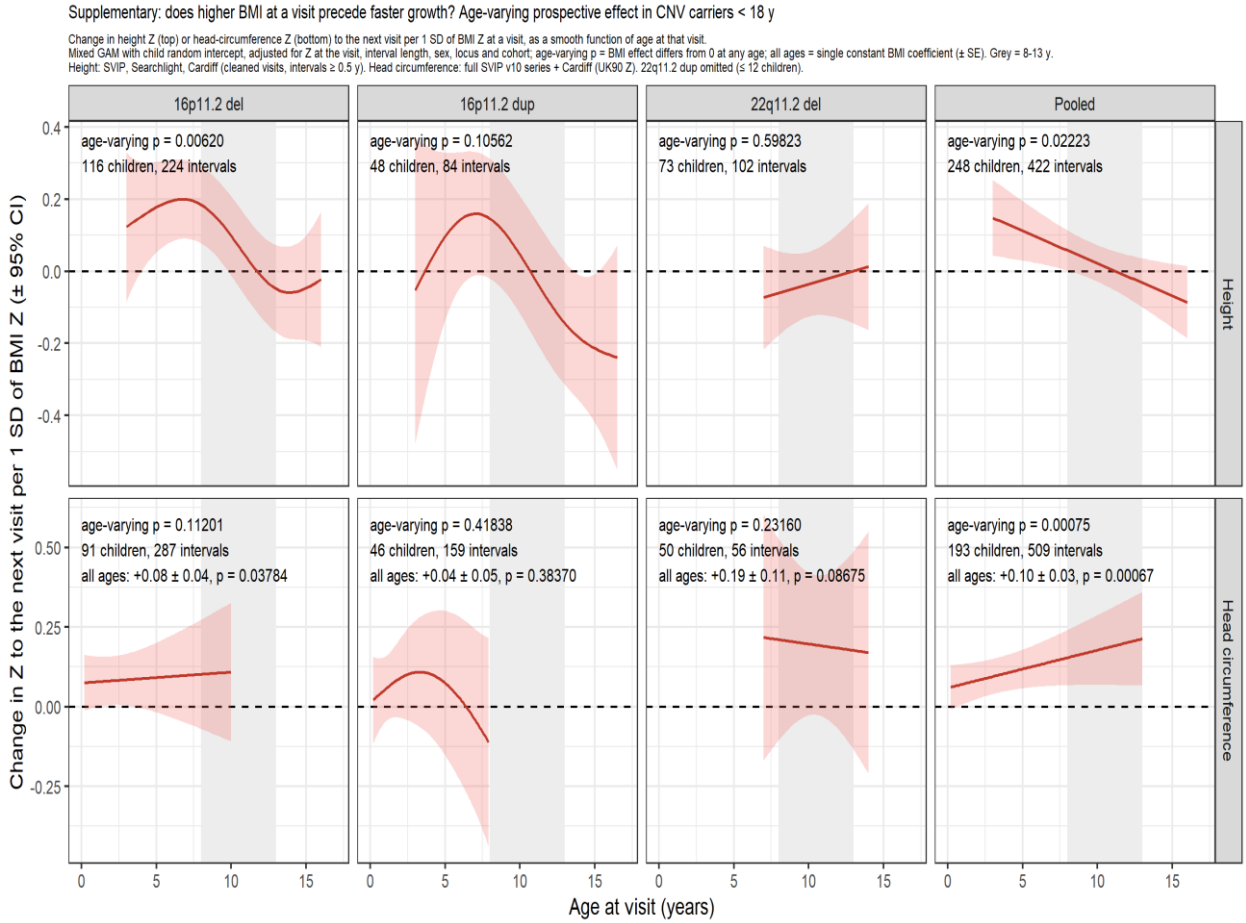
Penn CNB[13,14] accuracy and speed scores were standardized against G2MH non-carriers (speed as the negative z of median correct response time); totals are the mean of subtest z scores over at least five subtests, and BMI was taken within three years of the battery. G2MH non-carriers with CNB data have no BMI, so no control contrast is available.

Supplementary Figures

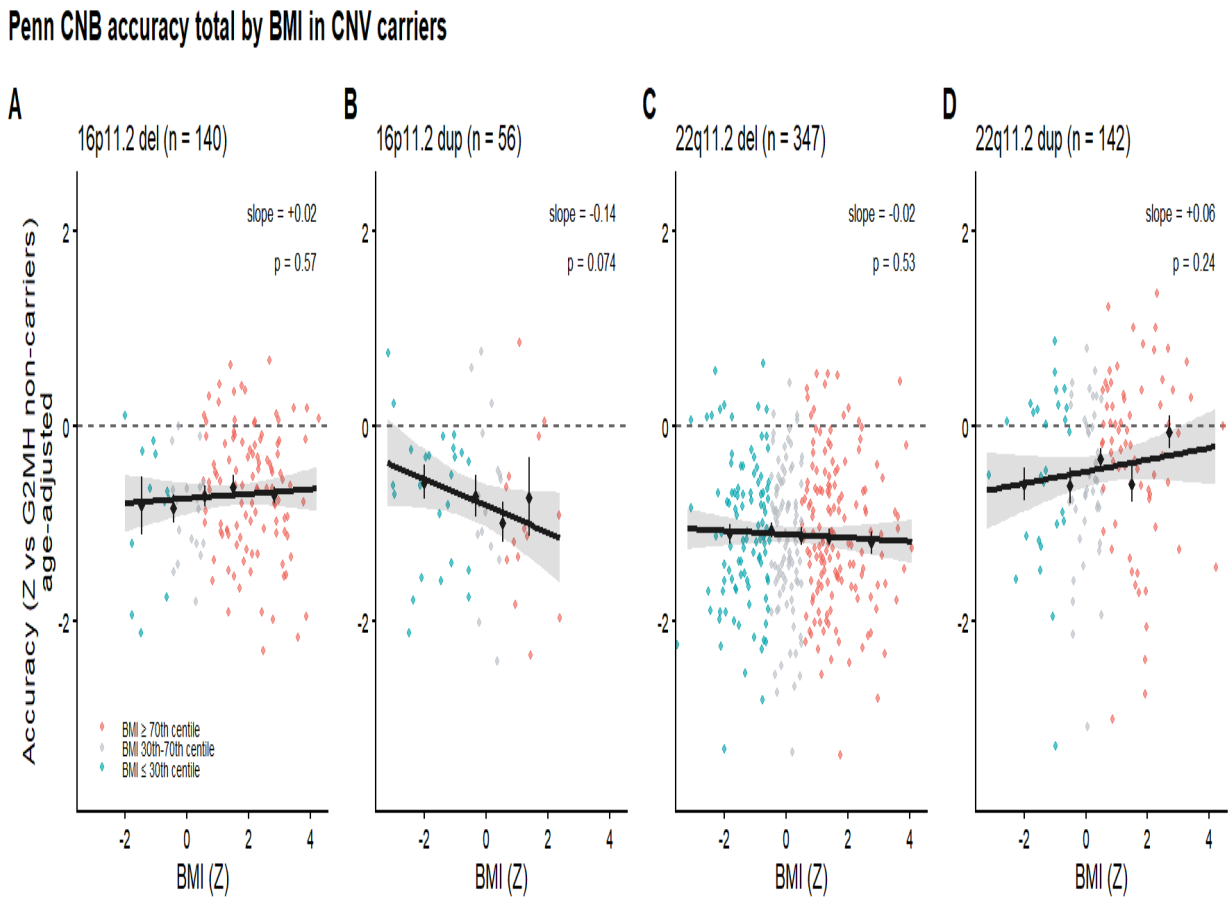
Supplementary Fig. 1 | Prospective, age-varying effect of body mass on subsequent growth in CNV carriers under 18 years.

Change in height z (WHO; top row) or head-circumference z (UK90; bottom row) from one visit to the next, per 1 s.d. of BMI z at the first visit, as a smooth function of age at that visit (line, 95% CI). Columns: 16p11.2 deletion, 16p11.2 duplication, 22q11.2

deletion and all carriers pooled (including 22q11.2 duplication children). Mixed GAM with a child random intercept, adjusted for *z* at the visit, interval length, sex and (pooled) locus and cohort; the age-varying *P* tests whether the BMI effect differs from zero at any age; 'all ages' is the same model with a single constant BMI coefficient (+/- s.e.). Grey band, 8-13 years. Height: 116, 48, 73 and 248 children (224, 84, 102 and 422 intervals); head circumference: 91, 46, 50 and 193 children (287, 159, 56 and 509 intervals). Two-sided tests. Statistics in Supplementary Table 3.

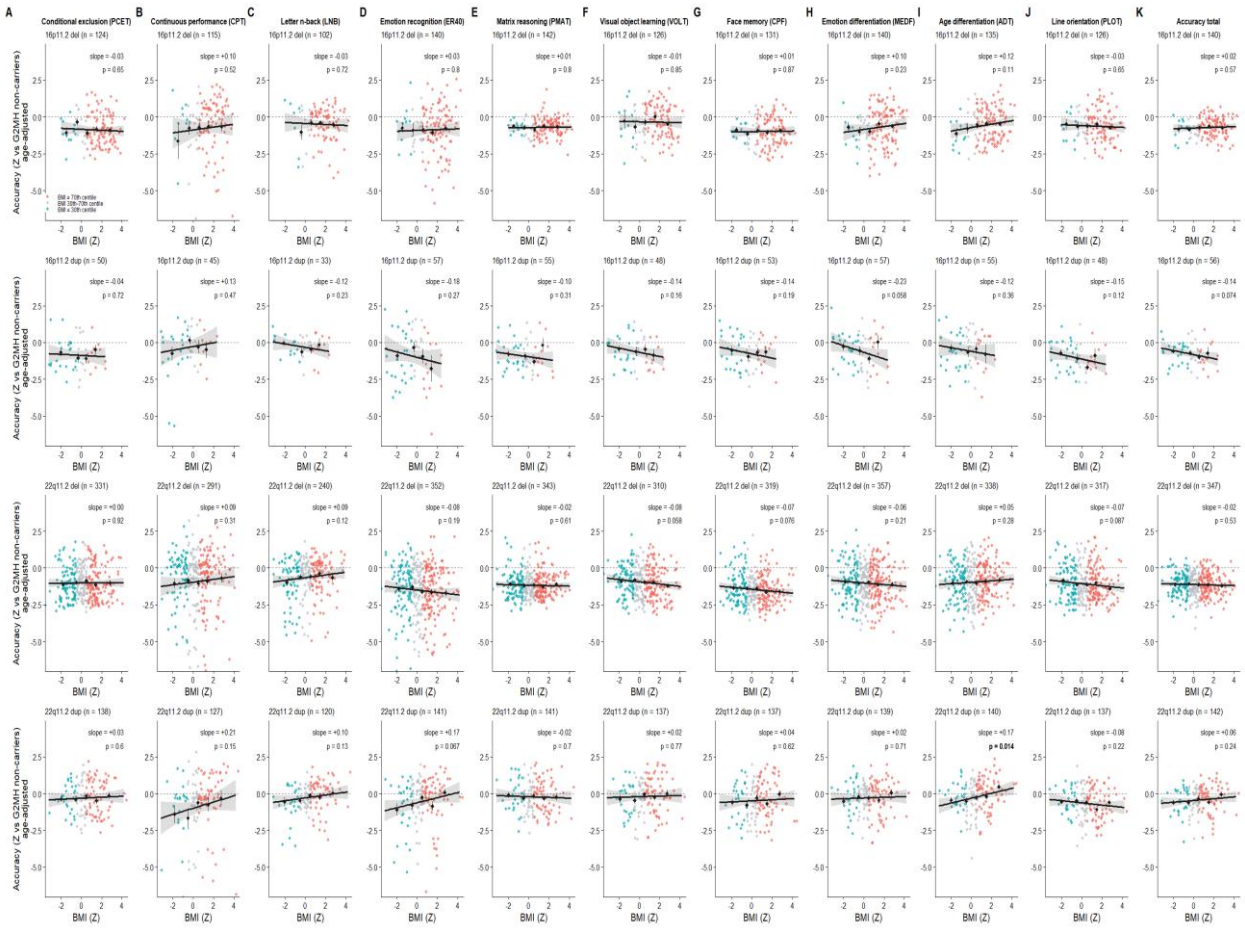


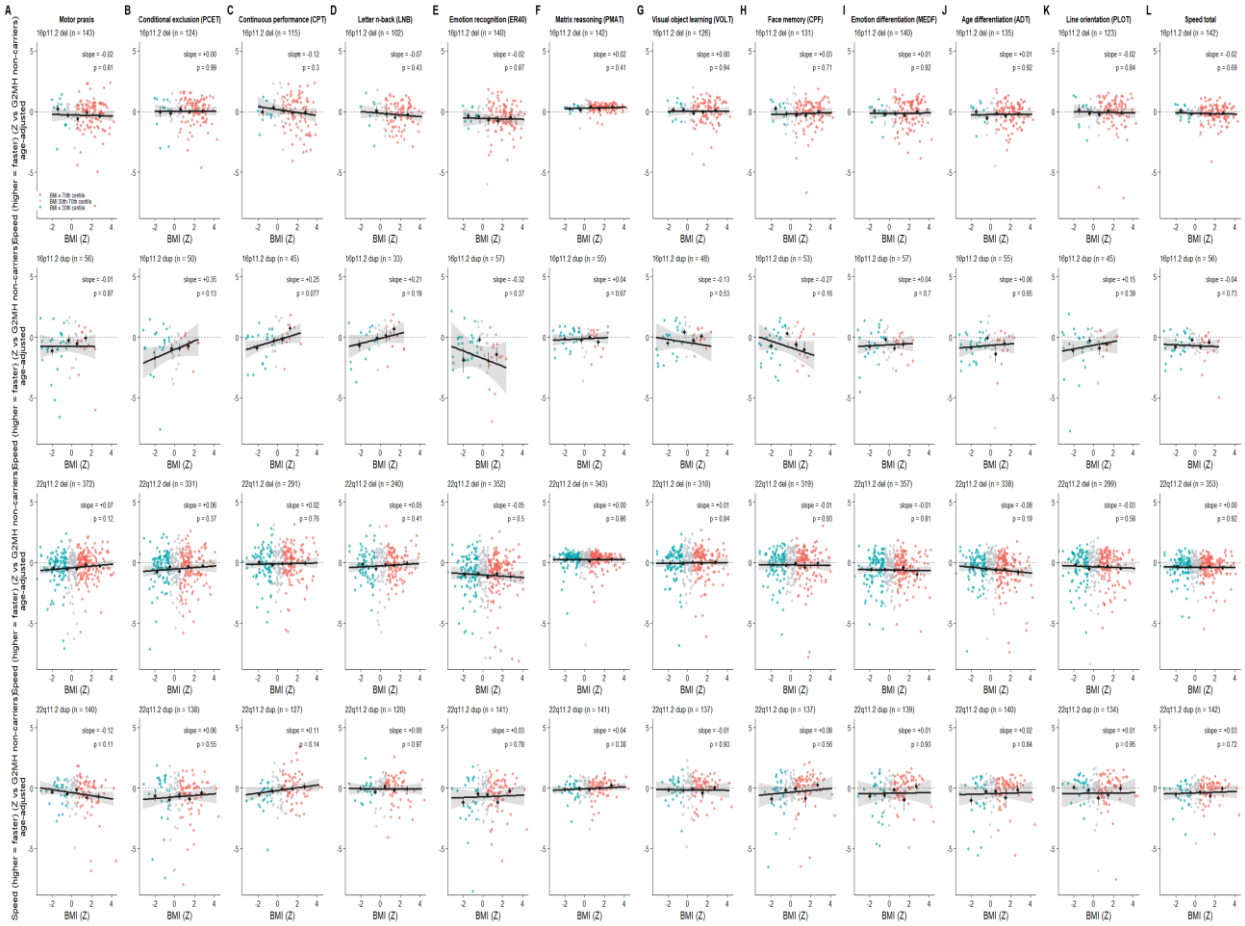
Supplementary Fig. 2 | Body mass and Penn CNB accuracy in CNV carriers. Penn Computerized Neurocognitive Battery accuracy total (mean *z* across subtests relative to G2MH non-carriers; at least five subtests), adjusted for age within group, against BMI *z* (measured within three years of the battery), in carriers of 16p11.2 deletion (**a**, *n* = 140), 16p11.2 duplication (**b**, *n* = 56), 22q11.2 deletion (**c**, *n* = 347) and 22q11.2 duplication (**d**, *n* = 142). Points coloured by BMI tail as in Fig. 1; black line and band, linear fit with 95% CI; diamonds, mean +/- s.e.m. in BMI bins. Printed, the BMI slope (s.d. per s.d.; GAM $y \sim s(\text{age}) + \text{BMI}$) and its two-sided *P* value. Heterogeneity of the slope across the four CNVs (Wald test of locus x BMI): $F = 2.02$ on 3 d.f., $P = 0.11$. G2MH non-carriers with CNB data have no BMI, so no control panel is shown.



Supplementary Fig. 3 | Body mass and Penn CNB subtest performance in CNV carriers.

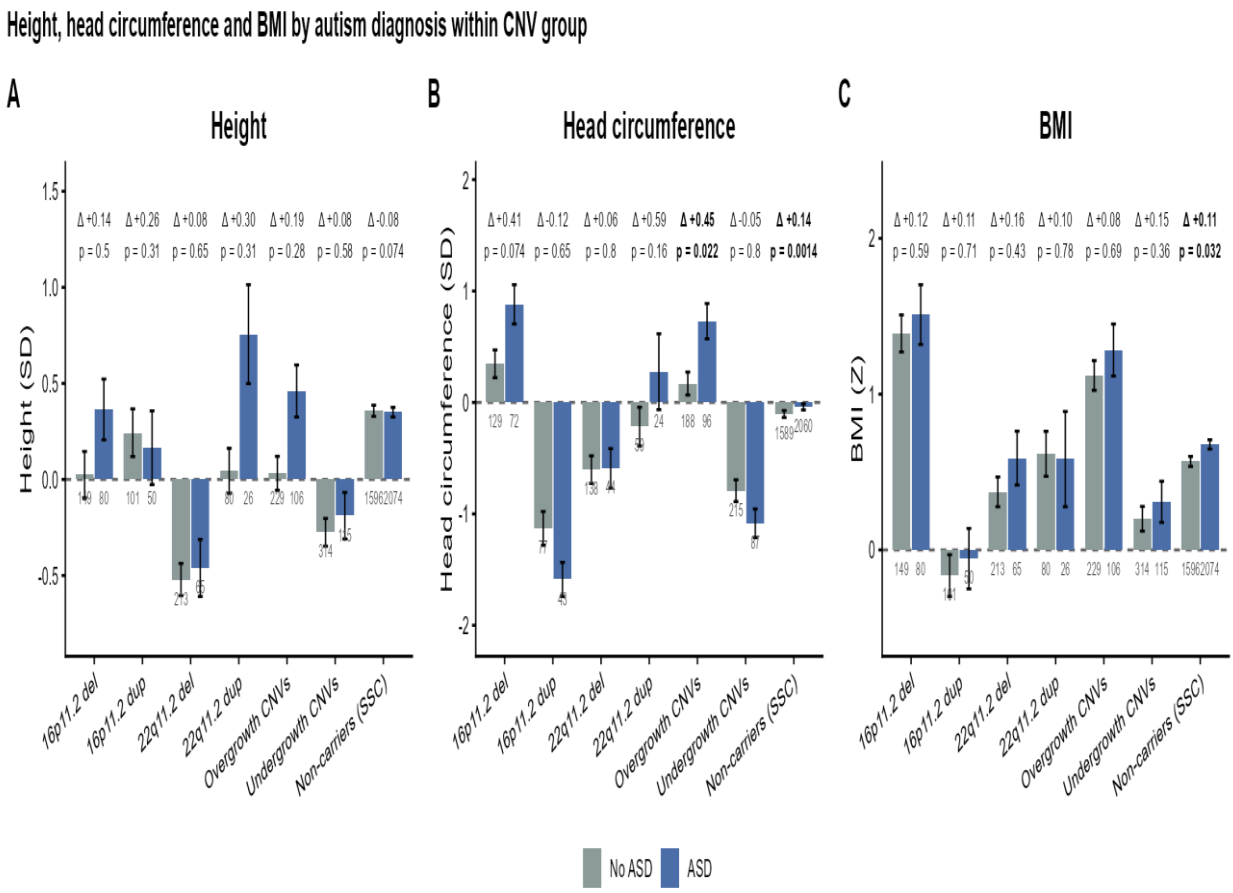
a, Accuracy (ten subtests and the accuracy total). **b**, Speed (eleven subtests and the speed total; speed is the negative *z* of median correct response time, higher = faster). Columns, CNB measures; rows, 16p11.2 deletion, 16p11.2 duplication, 22q11.2 deletion and 22q11.2 duplication carriers (G2MH). Each panel plots the age-adjusted score against BMI *z* with the conventions of Supplementary Fig. 2 and prints the BMI slope (s.d. per s.d.) and its two-sided *P* value (bold, $P < 0.05$). *n* (accuracy total / speed total): 140/142, 56/56, 347/353, 142/142. No subtest or total shows significant heterogeneity of the slope across CNVs (accuracy total $P = 0.11$; speed total $P = 0.94$; all subtests $P >= 0.086$).


A
Conditional exclusion (PCET)
B
Continuous performance (CPT)
C
Letter n-back (LNB)
D
Emotion recognition (ER40)
E
Matrix reasoning (PMAT)
F
Visual object learning (VOLT)
G
Face memory (CPF)
H
Emotion differentiation (MEDF)
I
Age differentiation (ADT)
J
Line orientation (PLOT)
K
Accuracy total
16p11.2 del (n = 124)
16p11.2 del (n = 115)
16p11.2 del (n = 102)
16p11.2 del (n = 140)
16p11.2 del (n = 142)
16p11.2 del (n = 126)
16p11.2 del (n = 131)
16p11.2 del (n = 140)
16p11.2 del (n = 135)
16p11.2 del (n = 126)
16p11.2 del (n = 140)
16p11.2 dup (n = 50)
16p11.2 dup (n = 45)
16p11.2 dup (n = 33)
16p11.2 dup (n = 57)
16p11.2 dup (n = 55)
16p11.2 dup (n = 48)
16p11.2 dup (n = 53)
16p11.2 dup (n = 57)
16p11.2 dup (n = 55)
16p11.2 dup (n = 48)
16p11.2 dup (n = 56)
22q11.2 del (n = 331)
22q11.2 del (n = 291)
22q11.2 del (n = 240)
22q11.2 del (n = 352)
22q11.2 del (n = 343)
22q11.2 del (n = 310)
22q11.2 del (n = 319)
22q11.2 del (n = 357)
22q11.2 del (n = 338)
22q11.2 del (n = 317)
22q11.2 del (n = 347)
22q11.2 dup (n = 138)
22q11.2 dup (n = 127)
22q11.2 dup (n = 120)
22q11.2 dup (n = 141)
22q11.2 dup (n = 141)
22q11.2 dup (n = 137)
22q11.2 dup (n = 137)
22q11.2 dup (n = 139)
22q11.2 dup (n = 140)
22q11.2 dup (n = 137)
22q11.2 dup (n = 142)
Accuracy (Z vs G2MH non-carriers) age-adjusted
BMI (Z)
BMI > 70th centile
BMI 30th-70th centile
BMI < 30th centile



Supplementary Fig. 4 | Height, head circumference and BMI by autism diagnosis within CNV group.

Mean +/- s.e.m. of height *z* (**a**), head-circumference *z* (**b**) and BMI *z* (**c**) in carriers with (blue) versus without (grey) an autism diagnosis, for each CNV, the pooled overgrowth (16p11.2 deletion + 22q11.2 duplication) and undergrowth (16p11.2 duplication + 22q11.2 deletion) groups and, as the non-carrier comparison, SSC probands with autism versus SSC designated siblings. Numbers under the bars are *n*. Printed above each group is the age- and sex-adjusted autism effect (delta, s.d.; GAM $y \sim$ autism + s(age) + sex, plus cohort and, for pooled groups, locus) with its two-sided *P* value (bold, $P < 0.05$). Carriers with autism are mostly children and carriers without include adult relatives; the age smooth addresses this and the head-circumference effect holds within children. The difference between each carrier group's autism effect and the proband-sibling difference (autism x carrier interaction), and the same contrasts with polygenic scores as covariates, are in Supplementary Tables 7 and 8.



Supplementary Tables

Supplementary Tables 1-10 are provided as an Excel workbook (Supplementary_Tables.xlsx; a 'notes' sheet defines every column). Effect sizes are in s.d. of outcome per s.d. of BMI unless stated; all *P* values are two-sided and unadjusted.

Table	Title	Source
Supplementary Table 1	Height and head circumference: within-group BMI slope and age x BMI, and CNV x BMI and CNV x age x BMI versus unaffected relatives and versus autistic SSC probands, per locus and pooled	supp_table_S1_growth.csv
Supplementary Table 2	SRS, SCQ and Penn CNB accuracy and speed: the same quantities (CNB within carriers only)	supp_table_S2_cognition.csv

Table	Title	Source
Supplementary Table 3	Prospective effect of BMI on subsequent height and head-circumference growth in carriers under 18 years: age-varying P, all-ages coefficient (estimate, s.e., t, P) and age-window coefficients	supp_table_S3_prospective.csv
Supplementary Table 4	Main-figure tests within strata of sex and of childhood obesity, adult BMI, height and autism polygenic scores (carriers and controls stratified alike)	supp_table_S4_sex_pgs_strata.csv
Supplementary Table 5	BMI effects with sex and polygenic scores as covariates (confounding check; same individuals in every row)	supp_table_S5_pgs_adjusted.csv
Supplementary Table 6	Power to detect modifier x BMI interactions (sex and each polygenic score) and the observed interactions	supp_table_S6_interaction_power.csv
Supplementary Table 7	Height, head circumference and BMI by autism diagnosis within each CNV group, the pooled groups and SSC probands versus siblings: means, n, age- and sex-adjusted autism effect, and the autism x	supp_table_S7_asd_by_cnv.csv

Table	Title	Source
	carrier interaction versus SSC non-carriers	
Supplementary Table 8	Supplementary Table 7 contrasts and interactions in the subset with polygenic scores, unadjusted and adjusted for each score and for all four	supp_table_S8_asd_by_cnv_pgs_adjusted.csv
Supplementary Table 9	Medication sensitivity: prevalence of current psychotropic use and within-carrier BMI slopes on height, head circumference, SRS, SCQ and CNB unadjusted, adjusted for medication classes, in unmedicated carriers, and adjusted for classes plus autism	medication_sensitivity.csv; medication_prevalence.csv
Supplementary Table 10	BMI slope on SRS and SCQ by group and autism status: SSC siblings, SSC parents, autistic SSC probands, each CNV and all CNVs, for all carriers, carriers without an autism diagnosis and all carriers with autism diagnosis as a covariate (slope, s.e., P, 95% CI, and the difference from unaffected relatives with its	cognition_bmi_slopes.csv (24_cognition_bmi_slopes.R)

Table	Title	Source
	P); the contrasts against autistic probands for all carriers are in Supplementary Table 2	

Supplementary Table 9 (abridged) | Age-adjusted BMI slope (s.d. of trait per s.d. of BMI; estimate (s.e.), P) by medication model, G2MH carriers.

Trait	Group	n	Unadjusted	+ medication classes	No psychotropic (n)
Height	16p11.2 del	89	+0.31 (0.11), p = 0.0048	+0.32 (0.11), p = 0.0043	+0.27 (0.12), p = 0.027 (72)
Height	16p11.2 dup	49	+0.07 (0.12), p = 0.6	+0.05 (0.13), p = 0.69	+0.05 (0.15), p = 0.75 (33)
Height	22q11.2 del	252	+0.05 (0.05), p = 0.34	+0.05 (0.05), p = 0.4	+0.04 (0.06), p = 0.53 (163)
Height	22q11.2 dup	90	+0.12 (0.09), p = 0.2	+0.04 (0.10), p = 0.65	+0.01 (0.11), p = 0.93 (67)
Height	All CNVs	480	+0.12 (0.04), p = 0.006	+0.11 (0.04), p = 0.0092	+0.09 (0.05), p = 0.059 (335)
Head circumference	16p11.2 del	67	+0.47 (0.09), p = 5.6e-06	+0.48 (0.09), p = 2.3e-06	+0.50 (0.11), p = 3.9e-05 (52)
Head circumference	16p11.2 dup	36	+0.07 (0.16), p = 0.69	+0.07 (0.17), p = 0.67	+0.02 (0.19), p = 0.92 (26)
Head circumference	22q11.2 del	142	+0.34 (0.08), p = 1.6e-05	+0.33 (0.08), p = 4.7e-05	+0.30 (0.10), p = 0.0031 (98)
Head circumference	22q11.2 dup	65	+0.43 (0.12), p = 8.4e-04	+0.39 (0.14), p = 0.0063	+0.32 (0.16), p = 0.06 (45)
Head circumference	All CNVs	310	+0.37 (0.05), p = 4.8e-12	+0.36 (0.05), p = 1.9e-11	+0.33 (0.06), p = 6.5e-07 (221)
SRS	16p11.2 del	69	+0.09 (0.22), p = 0.69	+0.09 (0.23), p = 0.7	+0.05 (0.28), p = 0.87 (54)
SRS	16p11.2 dup	30	+0.41 (0.32), p = 0.22	+0.30 (0.34), p = 0.39	n/a

Trait	Group	n	Unadjusted	+ medication classes	No psychotropic (n)
SRS	22q11.2 del	183	+0.27 (0.11), p = 0.013	+0.23 (0.10), p = 0.027	+0.29 (0.13), p = 0.026 (115)
SRS	22q11.2 dup	60	+0.09 (0.24), p = 0.71	+0.10 (0.27), p = 0.7	+0.06 (0.34), p = 0.86 (42)
SRS	All CNVs	342	+0.23 (0.09), p = 0.011	+0.19 (0.09), p = 0.03	+0.22 (0.11), p = 0.044 (229)
SCQ	16p11.2 del	52	+0.45 (0.35), p = 0.21	+0.26 (0.35), p = 0.45	+0.13 (0.36), p = 0.73 (44)
SCQ	22q11.2 del	146	+0.37 (0.16), p = 0.025	+0.37 (0.16), p = 0.024	+0.41 (0.19), p = 0.04 (90)
SCQ	22q11.2 dup	45	+0.38 (0.37), p = 0.32	+0.55 (0.43), p = 0.21	+0.82 (0.55), p = 0.15 (29)
SCQ	All CNVs	261	+0.35 (0.14), p = 0.011	+0.37 (0.14), p = 0.0068	+0.38 (0.17), p = 0.023 (176)
CNB accuracy	16p11.2 del	133	+0.03 (0.04), p = 0.4	+0.04 (0.04), p = 0.34	+0.04 (0.05), p = 0.38 (107)
CNB accuracy	16p11.2 dup	50	-0.06 (0.08), p = 0.47	-0.05 (0.08), p = 0.53	-0.06 (0.09), p = 0.48 (43)
CNB accuracy	22q11.2 del	315	-0.02 (0.03), p = 0.57	-0.02 (0.03), p = 0.54	+0.03 (0.04), p = 0.43 (203)
CNB accuracy	22q11.2 dup	125	+0.06 (0.06), p = 0.32	+0.07 (0.06), p = 0.21	+0.01 (0.07), p = 0.84 (89)
CNB accuracy	All CNVs	623	+0.00 (0.02), p = 0.88	+0.01 (0.02), p = 0.82	+0.02 (0.03), p = 0.39 (442)
CNB speed	16p11.2 del	135	-0.01 (0.05), p = 0.75	-0.01 (0.05), p = 0.86	+0.01 (0.05), p = 0.81 (108)
CNB speed	16p11.2 dup	50	+0.00 (0.11), p = 0.99	-0.01 (0.11), p = 0.94	-0.03 (0.12), p = 0.82 (43)
CNB speed	22q11.2 del	321	-0.01 (0.03), p = 0.87	-0.01 (0.03), p = 0.68	-0.01 (0.04), p = 0.85 (208)

Trait	Group	n	Unadjusted	+ medication classes	No psychotropic (n)
CNB speed	22q11.2 dup	125	+0.04 (0.09), p = 0.65	+0.05 (0.09), p = 0.6	+0.08 (0.08), p = 0.32 (89)
CNB speed	All CNVs	631	-0.00 (0.03), p = 0.91	-0.00 (0.03), p = 0.87	+0.00 (0.03), p = 0.94 (448)

n, carriers with medication information; % psychotropic, current use. 'No psychotropic' is the slope in carriers on no current psychotropic (n in brackets; omitted where n < 20). The full table, including medication prevalence by class, the weight-gain and weight-loss groupings and the model with medication classes plus autism diagnosis, is sheet 'Supplementary Table 9' of the workbook.

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